ThaiSLM — Latent space v8 (a Thai-only encoder: what it learned, what it no longer inherits)

All numbers: the same held-out queries, the same train-only bank, the same v8 labels, for every encoder (`artifacts/reports/isolated_v8.json`, `python main.py train-encoder --report …`, `lab.ipynb` §3). Encoders: **zero-shot** = Uni-Sign (CSL-News → WLASL) without Thai training; **v7** = Uni-Sign fine-tuned on Thai; **v8** = three small encoders trained from scratch on Thai data only (ensemble).

1. The ASL/CSL prior — measured, then removed

v7 reported that 39 % of its errors were "inherited" from the zero-shot model, and asked whether that was a prior bias or just signs that look alike. A model that never saw ASL/CSL answers it: if the overlap were only look-alike signs, a Thai-only model would share the same wrong answers just as often.

	v7	v8a (one member)	v8c	v8 ensemble
test errors (same 2,000 held-out queries)	1,839	1,853	1,830	**1,780**
errors that are the zero-shot model's same wrong answer	**41 %**	17 %	16 %	**20 %**
… per query	0.261	0.107	0.100	0.120

The overlap halves, so most of v7's inherited errors **were** the prior. What remains (~1 in 5) is shared because those signs genuinely look alike in pose space. v8 removes the bias without paying for it in errors.

2. Is the space organised by sign, not by signer?

5-NN neighbourhood	zero-shot	v7	v8
same concept	0.278	0.354	**0.392**
same signer	0.306	0.254	0.258
same dataset	0.709	0.650	0.668

v8 groups clips by sign more than v7 (+0.04). Signer and dataset are about as strong as in v7: each source still records its own signing style, which is why one-source words stay hard.

3. Signers per word decides recognition — v8 uses depth better

R@1 by people signing the word in the bank	1	2	3–4	5–9	10+
zero-shot	0.116	0.113	0.366	0.436	0.789
v7	**0.159**	0.217	0.422	0.414	0.855
v8	0.130	**0.255**	**0.530**	**0.543**	**0.859**

v8 is better for every word that at least two people sign (+4 to +13 points), and worse only for one-signer words, where the pre-training's generic motion knowledge helped v7. That is why the v8 "daily" vocabulary keeps only words with ≥ 2 signers (plus the curated conversation list).

4. Continuous windows — robustness to boundaries

The first v8 encoder was brittle when a word window cut from a real sentence included part of the neighbouring signs (word recall on the unseen signer's sentence windows fell from 0.56 to 0.48 when the boundaries moved ±25 %; v7 0.52 → 0.52 on the tune sentences). Training with **co-articulation context** (other signs glued before / after, v8c) and averaging three members fixed it:

tune sentence windows, conversation vocabulary, R@1	aligned	boundaries moved ±25 %
v7	0.520	0.520
v8a	0.549	0.480
v8c (+ context)	0.484	0.458
v8 ensemble	**0.597**	**0.568**

5. Not-a-sign

Null-vs-sign AUC on the TSL51 researcher: zero-shot 0.82, v7 0.70, v8 0.74. As in v7, continuous spotting decides "not a sign" with the tagger and the transition filter, not with this score.

6. What would change the space most

More people per everyday word. The depth curve (§3) is still the strongest effect in the data: a word recorded by 3–4 people is recognised four times as often as a one-person word. The YouTube harvest of v7 plus the v8 audit gives 525 clean clips of 23 new people; the next gain is more such people for the 1,366 daily words, not a larger model.